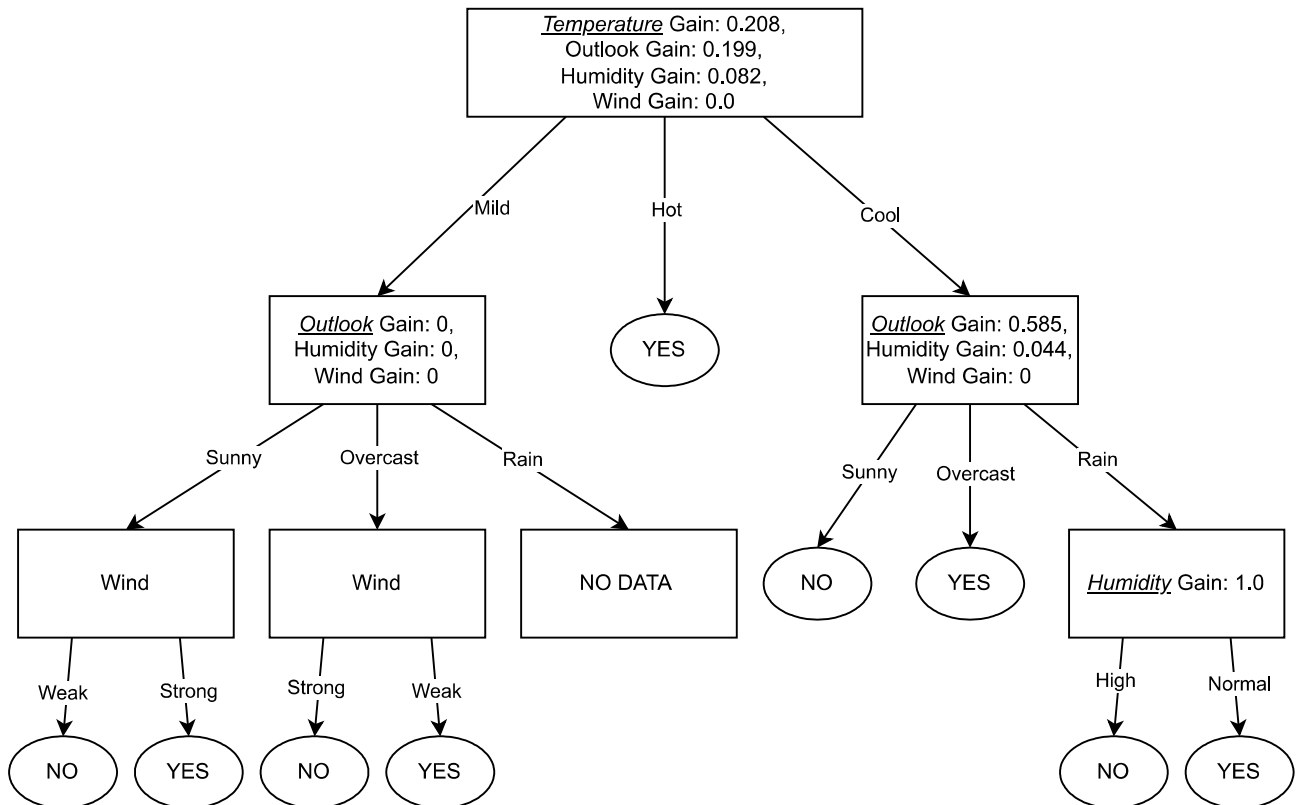




Attribute: OUTLOOK

- Overcast (3 Yes, 1 No): Entropy = 0.8113
- Sunny (1 Yes, 4 No): Entropy = 0.7219
- Rain (2 Yes, 1 No): Entropy = 0.9183
- > Gain(S, Outlook) = 0.1992

Attribute: TEMPERATURE

- Hot (2 Yes, 0 No): Entropy = 0
- Mild (2 Yes, 2 No): Entropy = 1
- Cool (2 Yes, 4 No): Entropy = 0.9183
- > Gain(S, Temperature) = 0.2075

Attribute: HUMIDITY

- Normal (4 Yes, 2 No): Entropy = 0.9183
- High (2 Yes, 4 No): Entropy = 0.9183
- > Gain(S, Humidity) = 0.0817

Attribute: WIND

- Weak (3 Yes, 3 No): Entropy = 1
- Strong (3 Yes, 3 No): Entropy = 1
- > Gain(S, Wind) = 0.0000

Task 1.2

Accuracy:

- 1.1 DT: 100%
- Given DT: 100%

Task 1.3

Both DTs achieved 100% accuracy, but the given DT is smaller and less complex than the one from Task 1.1, so I would prefer this one.

This shows that the ID3 algorithm is greedy and just looks at the best choice at that specific moment without further, which causes it to have more complex DTs than necessary.